

Elbow Anatomy and Biomechanics

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Elbow anatomy is complex, but knowledge of its intricate structural makeup is crucial to understanding athletic injuries in this area. Those who provide care for athletes, particularly athletes who participate in overhead-throwing sports, must have the ability to accurately diagnose and treat all forms of elbow injury and pathology. Accurate diagnosis and treatment require a thorough understanding of the interplay between the anatomy and function of the elbow joint.¹

ANATOMY

Elbow stability is afforded by both static and dynamic constraints.^{1,2} The primary static constraints include the ulnohumeral articulation, the anterior bundle of the ulnar collateral ligament (UCL), and the lateral collateral ligament (LCL) complex. Secondary static constraints include the common extensor and flexor pronator tendon origins, the joint capsule, and the radiocapitellar articulation.^{2,3} Dynamic stability is conferred by surrounding musculature that crosses the elbow and stabilizes the joint secondarily during active contraction, including the brachialis, biceps, triceps, flexor pronator mass, brachioradialis, extensor carpi radialis brevis, and longus.

Osseous Anatomy

The articular surfaces of the elbow joint consist of the distal humerus, radial head, and proximal ulna. The joint itself has been referred to as a trochleoginglymoid joint and comprises the ulnohumeral joint (hinged or ginglymoid joint) and the radiocapitellar articulation (trochoid).^{3,4}

The congruent articulation of the olecranon with the olecranon fossa is pivotal for elbow stability at terminal extension and greater than 120 degrees of flexion.^{1,5} In maximal elbow flexion, the coronoid process and radial head are locked into their respective fossae of the distal humerus, thereby imparting maximal stability.⁶ The distal humeral shaft has both medial and lateral supracondylar columns, as well as condyles. On the medial side is the spool-like trochlea, whereas on the lateral side is the spherical capitellum.⁷ The articular surface of the radial head articulates with the capitellum, whereas the radial head articulates with the sigmoid notch of the proximal ulna.⁷ Distally the biceps tendon inserts on the extraarticular radial tuberosity, whereas the triceps tendon inserts on the tip of the olecranon posteriorly.

The normal valgus orientation of the forearm relative to the arm varies as a function of both age and sex. It is smaller in children and averages 3 to 4 degrees more in females.^{8,9} The carrying angle represents the orientation of the forearm in reference to the humerus when the elbow is in full extension as viewed from anterior to posterior.¹⁰ It is formed by the valgus tilt of the axis of rotation (humeral articulation) and the valgus orientation of the ulnar shaft in relation to the olecranon.^{11–13} The normal angle varies, averaging about 10 degrees in males and 13 degrees in females (Fig. 57.1).

Capsuloligamentous Restraints

The capsule of the elbow joint contains all three bony articulations and incorporates medial and lateral thickenings, which help form the primary ligamentous stabilizers.⁷ The distended capsule may hold up to about 30 mL of fluid in the adult.¹⁴ The majority of the stabilizing effects of the capsule are derived from its anterior portion when the elbow is in full extension.¹⁵ In this regard, the anterior capsule resists hyperextension of the elbow, valgus stress, and joint distraction. In contrast, the posterior capsule may resist joint hyperflexion and posterior directed stresses.

The UCL complex, also referred to as the medial collateral ligament, comprises both anterior and posterior bundles and a variable transverse element (Cooper ligament)^{16–18} (Fig. 57.2). The anterior bundle of the UCL (aUCL) is the primary ligamentous stabilizer to valgus stress on the elbow. It originates at the anteroinferior aspect of the medial epicondyle and inserts on the sublime tubercle of the medial coronoid process (new¹⁹). The aUCL is further divided into anterior and posterior bands. These two bands have a cam effect because the origin of the ligament is slightly posterior to the flexion-extension axis.¹⁷ The anterior band of the aUCL is taut in extension and relaxes with further flexion, whereas the posterior band becomes more taut with elbow flexion. The posterior bundle of the UCL represents a capsular thickening and is less well defined.²⁰ The transverse bundle does not have a large role in joint stability.¹⁷

The LCL complex is composed of multiple structures, including the radial collateral ligament (RCL), the lateral ulnar collateral ligament (LUCL), the annular ligament, and the accessory lateral collateral ligament (Fig. 57.3). The LCL complex is a key stabilizer for resisting varus stress and posterolateral instability.²¹ The RCL and LUCL originate from the isometric point on at the inferior

Abstract

Elbow anatomy is complex, but knowledge of its intricate structural makeup is crucial to understanding athletic injuries in the area of the elbow. Physicians who provide care for athletes, particularly athletes who participate in overhead-throwing sports, must have the ability to accurately diagnose and treat all forms of elbow injury and pathology. Accurate diagnosis and treatment require a thorough understanding of the interplay between the anatomy and function of the elbow joint, particularly for overhead-throwing athletes.

Keywords

elbow anatomy
elbow biomechanics
ulnar collateral ligament
throwing cycle
elbow ligaments

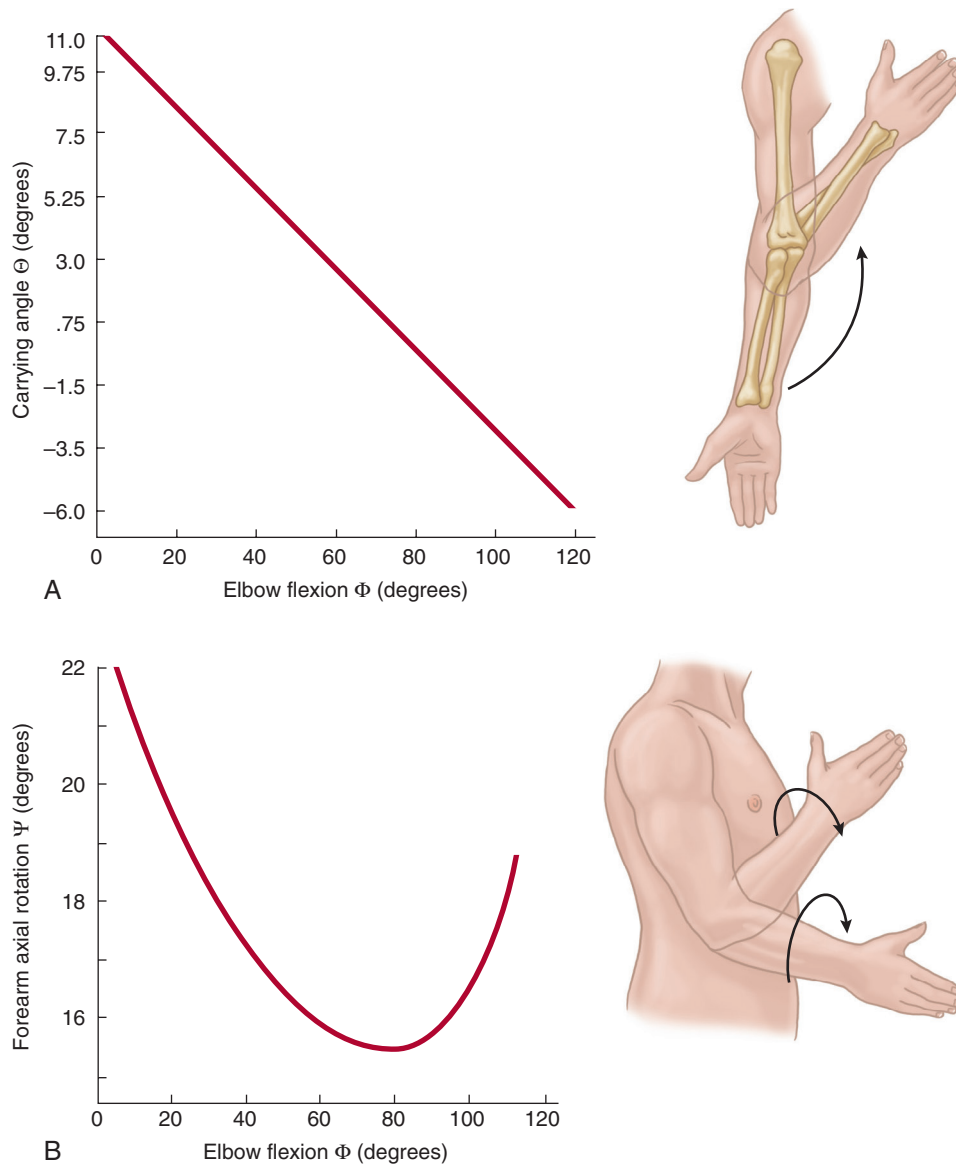


Fig. 57.1 (A) During elbow flexion, the carrying angle changes in a linear fashion from valgus to neutral or varus. (B) The ulna undergoes a slight axial rotation during flexion and extension. (Courtesy the Mayo Foundation.)

aspect of the lateral epicondyle. The RCL inserts into the annular ligament and stabilizes the radial head, whereas the LUCL attaches to the supinator crest of the proximal ulna and resists varus and posterolateral instability. The annular ligament forms a ring of tissue around the radial head, and inserts on the anterior and posterior aspects of the radial notch.

Elbow Musculature

Muscles crossing the elbow joint serve to control motion of the limb and provide dynamic stabilization to help the static structures. The primary groups acting on the joint include the elbow flexors (biceps, brachialis), elbow extensors (triceps), flexor pronator mass (resist valgus forces), and the mobile wad of three (brachioradialis, extensor carpi radialis longus, and extensor carpi radialis brevis—resist varus forces). The anconeus muscle may have a minor role in elbow extension but more importantly

is thought to function as a restraint to varus stress and posterolateral rotatory instability.⁶

ELBOW BIOMECHANICS

Joint Motion

Flexion-Extension

The normal arc of motion during flexion ranges from 0 degrees (full extension) to roughly 140 degrees. The axis of rotation (Fig. 57.4) for elbow flexion passes through the capitellum in line with the bottom of the trochlear sulcus.⁷

Pronation-Supination

Forearm rotation averages nearly 180 degrees through its arc. Pronation typically accounts for 80 to 85 degrees, and supination accounts for 85 to 90 degrees of motion. The axis of forearm

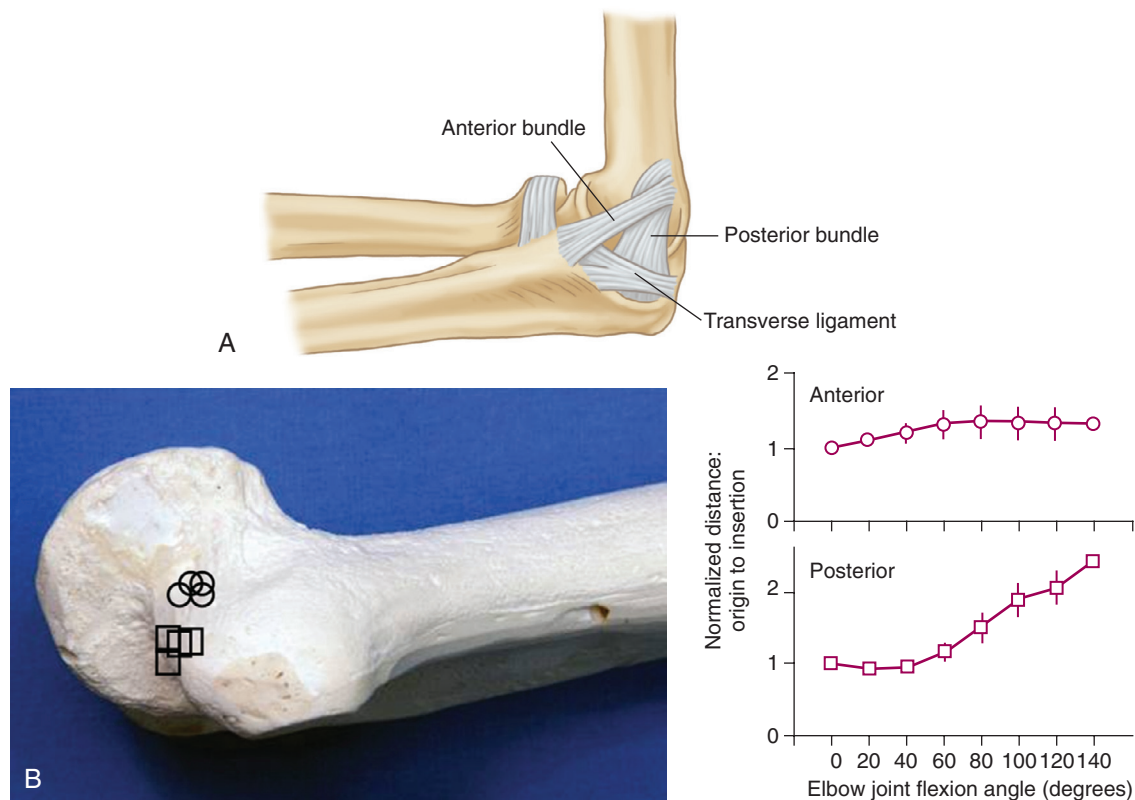


Fig. 57.2 (A) Anatomic distribution of the components of the medial collateral ligament complex. (B) The anterior bundle originates at the locus of the axis of rotation. However, the origin of the posterior bundle does not lie along the axis of rotation, and thus some change in the length of the posterior band is seen with changes in the elbow flexion angle. (Courtesy the Mayo Foundation.)

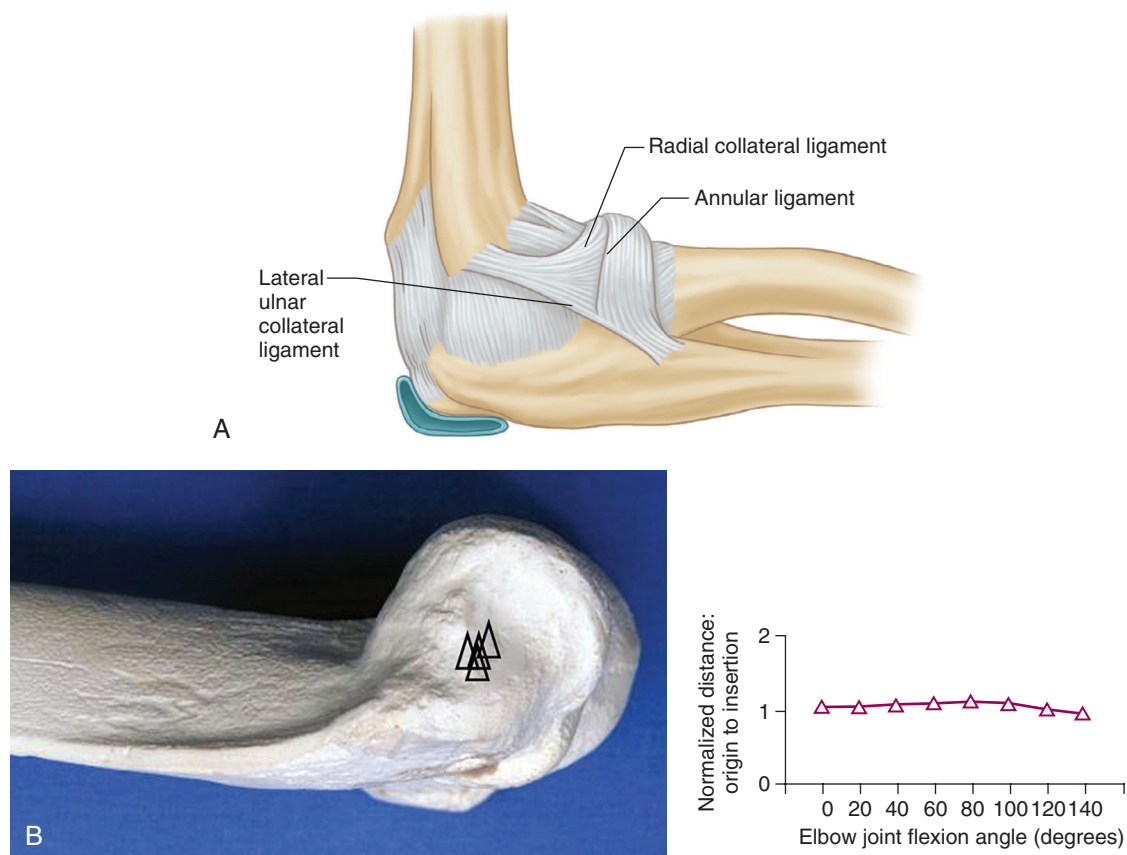


Fig. 57.3 (A) The lateral collateral ligament complex consists not only of the radial collateral ligament but also a lateral ulnar collateral ligament. (B) With flexion, no change occurs in the length of the radial collateral ligament complex, which suggests that the origin is at the axis of rotation. (Courtesy the Mayo Foundation.)



Fig. 57.4 The axis of rotation of the elbow is approximated by a line running through the middle of the lateral epicondyle and the center of the trochlea, emerging on the anterior inferior aspect of the medial epicondyle.

rotation runs from the center of the radial head to the fovea of the ulnar head.^{22,23} Ligamentous deficiency is to some degree offset or altered by forearm position; supination may help stabilize the UCL-deficient elbow, whereas pronation can add stability to an elbow with LCL insufficiency.^{24,25}

Motion Loss

There are a myriad of causes for elbow stiffness. A loss of flexion may be caused by bony impingement of the radial head or coronoid process against their respective fossae, loose bodies in the anterior capsule, or posterior capsular tightness.⁷ Extension may be limited because of posterior osteophyte impingement or anterior capsular tightness, although a mild flexion contracture is an adaptive variation of normal in the dominant throwing arm of many athletes.²⁶ Numerous studies have examined the question of what constitutes “functional range of motion.”²⁷ Morrey et al.²⁸ noted that most activities of daily living, such as grooming, feeding, and bathing, could be performed with a total elbow flexion arc of 100 degrees (30 to 130 degrees) and a forearm rotation arc of 100 degrees, with 50 degrees each of both pronation and supination. More recently, Sardelli et al.²⁹ noted that the use of cell phones requires greater elbow flexion, whereas use of computer keyboards requires more forearm pronation than may have been previously appreciated.

Joint Stability

Osteoarticular Stability

Distal humeral articular anatomy. The distal humerus is composed of two articular surfaces, named the capitellum and the trochlea. The capitellum articulates with radial head in the lateral aspect of the joint. The capitellar shape is hemispherical, accommodating the concave elliptical nature of the radial head. The trochlea is often described as spool-shaped. It angles slightly posteriorly to minimize posterior translation when engaged by the coronoid.³⁰

Coronoid process. The coronoid of the proximal ulna is critical to elbow stability.³¹ It is composed of anteromedial and anterolateral facets, a body and tip. Given that the tip of the



Fig. 57.5 Absence of 50% of the coronoid is estimated by drawing a line from the olecranon tip to the coronoid release. The line is parallel to the shaft at the 50% level.

coronoid is devoid of soft tissue attachments, fractures tend to be secondary to shear and/or axial loading.^{3,32} Located on the anteromedial facet is the sublime tubercle to which the aUCL inserts. Studies have shown that fractures involving more than 50% of the coronoid (Fig. 57.5) are associated with increased instability at the elbow in extension.^{3,33,34} In addition, in conjunction with the radial head, the coronoid is important for posterolateral stability.³⁵ Approximately 30% to 40% of the coronoid may be removed without consequent ulnohumeral instability; further resection of the coronoid process, particularly in the setting of concomitant fracture of the radial head, will consistently lead to posterior dislocation.³⁶ The coronoid is also important for varus stability in conjunction with an intact LCL complex.³⁷

Radial head. The radial head is an important secondary static stabilizer of the elbow, particularly under valgus load.^{16,38} In this regard, its importance is increased in the setting of UCL insufficiency (Fig. 57.6).³⁹ Posterolateral rotatory instability may manifest after isolated radial head resection, most likely because of diminished tension on the LCL complex or concurrent injury to the LUCL itself.⁴⁰ Furthermore, radial head deficiency in the setting of accompanying interosseous membrane disruption may lead to longitudinal instability, characterized by proximal radial

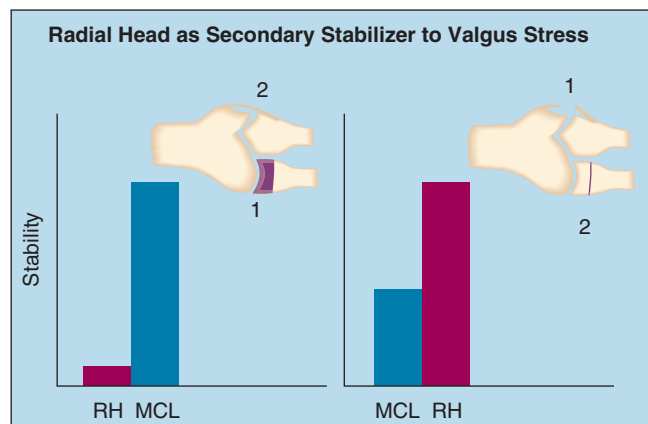


Fig. 57.6 Left, Removing the radial head (RH) (1) but leaving the medial collateral ligament (MCL) intact (2) results in no alteration in valgus stability. Right, When the MCL is removed first (1), the RH is observed to provide some resistance to valgus stress. When both constraints are released (2), the elbow is grossly unstable. (Courtesy the Mayo Foundation.)

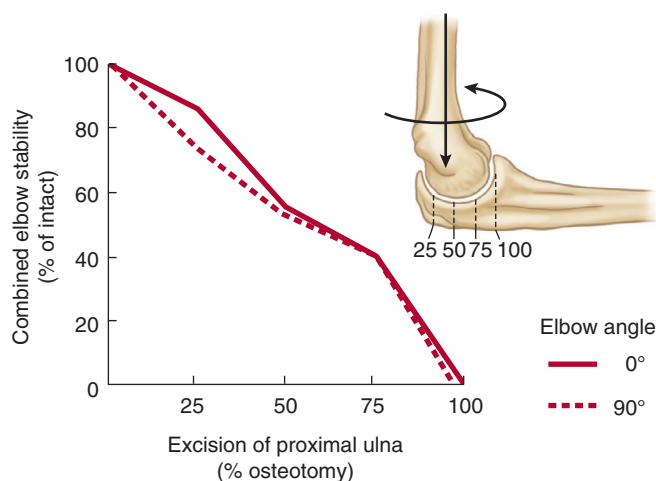


Fig. 57.7 With serial removal of the olecranon, a linear decrease in ulnohumeral stability is observed. Note the 50% loss with 50% resected. (Courtesy the Mayo Foundation.)

head migration, ulnar positive variance, and ulnar-sided wrist pain (Essex-Lopresti injury).

Olecranon process. The olecranon contributes to elbow stability by engaging with the olecranon fossa of the posterior distal humerus as the elbow extends. Not surprisingly, joint contact pressures increase with progressive resection of the olecranon (e.g., as may be performed for a comminuted olecranon fracture).⁴¹ Partial resection of the olecranon may increase coronal plane (varus/valgus) laxity (Fig. 57.7). It has been generically recommended that no more than 50% of the olecranon should be resected, although a recent in vitro study suggested that gross instability does not occur until resection levels greater than 87% are reached.⁴² Overhead-throwing athletes may not tolerate larger amounts of olecranon resection over as little as 3 mm because of amplified strain on the UCL with increasing angles of elbow flexion.⁴³

Soft Tissue Stability

Ulnar collateral ligament. The aUCL is the primary restraint to valgus force at the elbow.^{16,44,45} The anterior band of the aUCL is the primary restraint to valgus loads between 30 and 90 degrees of elbow flexion, and both the posterior and anterior bands function jointly as primary restraints at 120 degrees.⁴⁶ The aUCL also has a central band between the anterior and posterior bands that is essentially isometric throughout elbow flexion. With complete release of the aUCL, the elbow is maximally unstable in valgus at 70 degrees of flexion.^{3,47} The UCL provides a third of the valgus stability in full extension, compared with half of the stability at 90 degrees of flexion.³⁰

As previously mentioned, the UCL has a cam effect during elbow motion. This effect occurs because the ligament complex origin is posterior to the center of rotation, and thus ligament tension has been shown to change throughout the arc of motion, specifically increasing in length when the elbow moves from terminal extension to high angles of flexion.¹⁷

With an intact MCL, radial head excision does not impart valgus instability to the elbow. If the MCL is additionally injured, the joint becomes unstable.

Isolated injury to the posterior bundle of the MCL can lead to posteromedial rotatory instability.

Lateral collateral ligament. Much like its analogous structure at the medial elbow, the LCL complex is the structure primarily responsible for resisting varus stress and external rotation. In contrast to the UCL, the center of rotation passes through the origin of the LCL, ensuring enhanced isometry and the absence of a cam effect.¹⁷ With respect to the annular ligament, the anterior portion is taut during supination, whereas the posterior portion is taut in pronation. O'Driscoll et al.² stated that LCL injury is the initial injury in the spectrum of the pathology of elbow dislocations.

The individual anatomic structures of the LCL likely function together as a complex. With an intact annular ligament, both the LUCL and RCL need to be cut to produce gross coronal plane instability or posterolateral rotatory instability.⁴⁸ Individual transection of the annular ligament or LUCL produces only minor laxity.⁴⁹ In combined injuries with radial head deficiency and LCL complex insufficiency, instability with varus and external rotation forces is more pronounced. Replacement of the radial head restores partial stability, but simultaneous repair or reconstruction of the ligament complex is also necessary in this scenario (Fig. 57.8).⁵⁰ In the setting of posterolateral rotatory instability, loads applied across the flexed, supinated elbow can lead to posterior radial head subluxation, lateral elbow pain, and mechanical symptoms.⁵¹

Experimental in vitro data may be summarized by stating that the articular surfaces provide about 50% of elbow stability and the collateral ligaments provide the remaining 50%. When the elbow is in full extension, the anterior capsule contributes about 15% of the resistance to varus-valgus stress.^{10,16}

Muscular contributions and joint reactive forces. Muscles crossing the joint are dynamic stabilizers and are associated with corresponding joint reaction forces.⁷ For example, the flexor pronator musculature is consistently active during the throwing

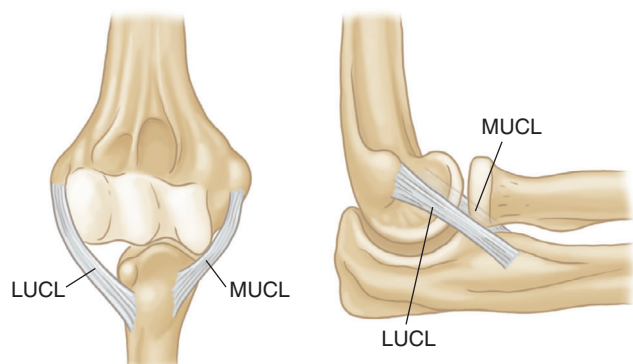


Fig. 57.8 The ulnohumeral joint is stabilized by both medial and lateral constraints, independent of the presence of the radial head. LUCL, Lateral ulnar collateral ligament; MUCL, medial ulnar collateral ligament. (Courtesy the Mayo Foundation.)

motion and contributes significantly to the dynamic stability of the elbow, especially if injury to the static restraints has been realized.¹ Evidence of the relative contribution of dynamic muscular stabilization is demonstrated in cadaveric sectioning studies, in which the ultimate tensile failure of the UCL (about 260 N) is less than the tensile forces experienced by the medial elbow during throwing.^{44,52} Therefore activation of these muscles provides protection of the static soft tissues. The flexor carpi ulnaris and portions of the flexor digitorum superficialis lie directly over the anterior bundle of the UCL.⁵³ With the arm relatively extended, approximately 40% of axial load across the elbow joint is experienced at the ulnohumeral joint, with almost 60% of the force transmitted across the radiocapitellar joint (Fig. 57.9).⁵⁴ Force transmission across the radial head is greatest in the 0- to 30-degree arc and is also larger with forearm pronation. Absence of the radial head results in higher forces at the ulnotrochlear joint, which is enhanced further in the setting of collateral ligament insufficiency.

The throwing cycle. Knowledge of the throwing cycle aids practitioners who evaluate and treat overhead-throwing athletes with elbow pathology. The overhead-throwing motion can be broken down into six phases: (1) windup, (2) early cocking, (3) late cocking, (4) acceleration, (5) deceleration, and (6) follow-through.^{1,55–57} Medial tensile forces and lateral compression forces are generated at the elbow during the late cocking and acceleration phases, whereas deceleration causes high shear stress, especially in the posterior compartment. Biomechanical testing has estimated valgus forces as high as 64 N · m at the elbow during the late cocking and early acceleration phases of throwing, with compressive forces of 500 N at the lateral radiocapitellar articulation as the elbow moves from approximately 110 to 20 degrees of flexion with angular velocities as high as 3000 degrees per second. This combination of large valgus loads and rapid elbow extension produces high tensile stress medially along the soft tissue restraints (UCL, flexor-pronator mass, medial epicondylar apophysis, and ulnar nerve), along with compression stress laterally (radiocapitellar joint) and shear stress in the posterior compartment (posteromedial tip of the olecranon, trochlea, and olecranon fossa). This phenomenon has been termed *valgus extension overload syndrome* and forms the basic pathophysiologic

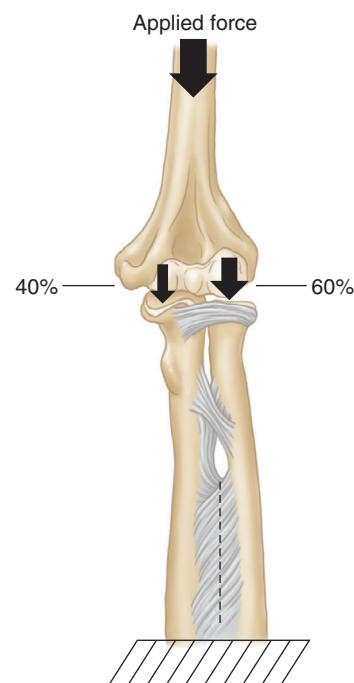


Fig. 57.9 The distributive forces with the elbow in full extension and an axial load placed at the wrist show that about 60% of the force goes across the radiocapitellar joint and 40% goes across the ulnohumeral joint; this does not change the section of the interosseous membrane. (Courtesy the Mayo Foundation.)

model explaining the mechanism for the most common elbow injuries in the overhead-throwing athlete.

More recently, the link between shoulder and elbow mechanics—as it relates to the overhead athlete—is becoming clearer. Elbow injuries requiring surgery in throwing athletes have been associated with decreased total shoulder rotation, as well as decreased shoulder forward flexion.⁵⁸ Furthermore, Anand et al. found that UCL insufficiency was associated with significant changes in contact area, contact pressure, and valgus laxity—during both relative flexion (such as the late cocking/early acceleration phase) as well as relative extension (deceleration phase) moments during the throwing motion arc.⁵⁹

The anterior bundle of the UCL is the primary restraint to valgus force of the elbow from 20 to 120 degrees of flexion and is subjected to near failure during the acceleration phase of the throwing motion. Repetitive loads near tensile failure result in microtrauma to the anterior band of the UCL and may eventually lead to ligament attenuation or rupture. Continued valgus and extension forces combined with subtle laxity are ingredients for further pathologic deterioration of the thrower's elbow.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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Citation:

Cain EL Jr, Dugas JR. History and examination of the thrower's elbow. *Clin Sports Med*. 2004;23(4):553–566.

Level of Evidence:

V

Summary:

This excellent review article highlights the basic history and examination of the elbow in overhead-throwing athletes.

Citation:

O'Driscoll SW, Jupiter JB, King GJ, et al. The unstable elbow. *Instr Course Lect.* 2001;50:89–102.

Level of Evidence:

V

Summary:

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Citation:

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Level of Evidence:

V

Summary:

This thorough review addresses the biomechanics and anatomy of the elbow.

Citation:

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Level of Evidence:

V

Summary:

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